Customer Feedback System – OpenAPI Stubs with Sample Payloads & Response Examples

This document provides concise OpenAPI (v3.0) stubs for each microservice along with sample request payloads and response examples. It also includes the deployment pipeline diagram for CI/CD.

Common Components

```
openapi: 3.0.3
info:
  title: Customer Feedback System APIs
  version: 0.2.0
servers:
  - url: https://api.example.com
components:
  securitySchemes:
    bearerAuth:
      type: http
      scheme: bearer
      bearerFormat: JWT
  schemas:
    ProblemDetails:
      type: object
      properties:
        type: { type: string }
        title: { type: string }
        status: { type: integer }
        traceId: { type: string }
        errors:
          type: object
          additionalProperties:
            type: array
            items: { type: string }
security:
  - bearerAuth: []
```

Auth Service

```
openapi: 3.0.3
info: { title: Auth Service, version: 0.2.0 }
paths:
  /api/auth/register:
    post:
      summary: Register a new user
      requestBody:
        required: true
        content:
          application/json:
            schema:
              type: object
              required: [email, password]
              properties:
                email: { type: string, format: email }
                password: { type: string, minLength: 8 }
                role: { type: string, enum: [Customer, StoreManager, Analyst, Admin] }
                storeId: { type: string, format: uuid }
            examples:
              sample:
                value:
                  email: manager@example.com
```

```
                password: "Str0ngP@ss!"
                role: StoreManager
                storeId: "2f1a8f7e-7e23-4e5d-8f12-8f7e23e58f12"
    responses:
      '201':
        description: Created
        content:
          application/json:
            examples:
              created:
                value:
                  id: "b3f2a0c7-1c2b-4bf6-9e21-5f3d4b1a2c9e"
                  email: manager@example.com
                  role: StoreManager
      '400':
        description: Validation error
        content:
          application/json:
            schema: { $ref: '#/components/schemas/ProblemDetails' }
            examples:
              invalid:
                value:
                  title: Validation Error
                  status: 400
                  errors:
                    password: ["Password must be at least 8 characters."]
/api/auth/login:
  post:
    summary: Login
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            required: [email, password]
            properties:
              email: { type: string, format: email }
              password: { type: string }
          examples:
            sample:
              value:
                email: manager@example.com
                password: "Str0ngP@ss!"
    responses:
      '200':
        description: Tokens
        content:
          application/json:
            schema:
              type: object
              properties:
                accessToken: { type: string }
                refreshToken: { type: string }
            examples:
              ok:
                value:
                  accessToken: "eyJhbGciOi..."
                  refreshToken: "def50200..."
      '401': { description: Unauthorized }
/api/auth/refresh:
  post:
    summary: Refresh token
    requestBody:
      required: true
      content:
        application/json:
          schema:
```

```
              type: object
              required: [refreshToken]
              properties: { refreshToken: { type: string } }
            examples:
              sample:
                value:
                  refreshToken: "def50200..."
      responses:
        '200':
          description: New tokens
          content:
            application/json:
              examples:
                ok:
                  value:
                    accessToken: "eyJhbGciOi..."
                    refreshToken: "def50201..."
        '401': { description: Unauthorized }
  /api/auth/me:
    get:
      summary: Get current user
      responses:
        '200':
          description: User profile
          content:
            application/json:
              examples:
                ok:
                  value:
                    id: "b3f2a0c7-1c2b-4bf6-9e21-5f3d4b1a2c9e"
                    email: manager@example.com
                    roles: ["StoreManager"]
        '401': { description: Unauthorized }
```

Feedback Service

```
openapi: 3.0.3
info: { title: Feedback Service, version: 0.2.0 }
paths:
  /api/feedback:
    post:
      summary: Submit feedback
      requestBody:
        required: true
        content:
          application/json:
            schema:
              type: object
              required: [storeId, rating]
              properties:
                storeId: { type: string, format: uuid }
                customerId: { type: string, format: uuid }
                rating: { type: integer, minimum: 1, maximum: 5 }
                comment: { type: string, maxLength: 2000 }
                categoryId: { type: string, format: uuid }
                metadata: { type: object }
            examples:
              sample:
                value:
                  storeId: "a9d3d7fd-6f47-4f3a-9a3d-3d7fd6f479a3"
                  customerId: "c1a2b3c4-d5e6-7890-abcd-ef1234567890"
                  rating: 4
                  comment: "Staff were helpful but checkout was slow."
                  categoryId: "0d9f1c0a-9b6e-4b3e-8b3c-7f1c0a9b6e4b"
                  metadata:
                    device: "web"
```

```
                locale: "en-IN"
    responses:
      '201':
        description: Created
        content:
          application/json:
            examples:
              created:
                value:
                  id: "9f4a1d3c-7b8e-4f0e-9a3c-1d3c7b8e4f0e"
                  createdAt: "2025-12-08T08:44:12Z"
      '400':
        description: Validation error
        content:
          application/json:
            examples:
              invalid:
                value:
                  title: "Validation Error"
                  status: 400
                  errors:
                    rating: ["Rating must be between 1 and 5."]
  get:
    summary: List feedback (paginated)
    parameters:
      - in: query
        name: storeId
        schema: { type: string, format: uuid }
      - in: query
        name: page
        schema: { type: integer, minimum: 1, default: 1 }
      - in: query
        name: size
        schema: { type: integer, minimum: 1, maximum: 100, default: 20 }
      - in: query
        name: from
        schema: { type: string, format: date-time }
      - in: query
        name: to
        schema: { type: string, format: date-time }
    responses:
      '200':
        description: Paged list
        content:
          application/json:
            examples:
              ok:
                value:
                  pageInfo:
                    page: 1
                    size: 20
                    totalItems: 135
                  items:
                    - id: "9f4a1d3c-7b8e-4f0e-9a3c-1d3c7b8e4f0e"
                      storeId: "a9d3d7fd-6f47-4f3a-9a3d-3d7fd6f479a3"
                      rating: 4
                      comment: "Helpful staff; slow checkout."
                      createdAt: "2025-12-08T08:44:12Z"
/api/feedback/{id}:
  get:
    summary: Get feedback by id
    parameters:
      - in: path
        name: id
        required: true
        schema: { type: string, format: uuid }
    responses:
      '200':
```

```
          description: Feedback
          content:
            application/json:
              examples:
                ok:
                  value:
                    id: "9f4a1d3c-7b8e-4f0e-9a3c-1d3c7b8e4f0e"
                    storeId: "a9d3d7fd-6f47-4f3a-9a3d-3d7fd6f479a3"
                    rating: 4
                    comment: "Helpful staff; slow checkout."
                    categoryId: "0d9f1c0a-9b6e-4b3e-8b3c-7f1c0a9b6e4b"
                    createdAt: "2025-12-08T08:44:12Z"
        '404': { description: Not found }
    put:
      summary: Update feedback
      requestBody:
        required: true
        content:
          application/json:
            examples:
              sample:
                value:
                  rating: 5
                  comment: "Team resolved the issue promptly."
      responses:
        '200':
          description: Updated
          content:
            application/json:
              examples:
                ok:
                  value:
                    id: "9f4a1d3c-7b8e-4f0e-9a3c-1d3c7b8e4f0e"
                    rating: 5
                    updatedAt: "2025-12-08T09:10:00Z"
        '403': { description: Forbidden }
    delete:
      summary: Delete feedback (soft)
      responses:
        '204': { description: Deleted }
```

Reporting Service

```
openapi: 3.0.3
info: { title: Reporting Service, version: 0.2.0 }
paths:
  /api/reporting/summary:
    get:
      summary: Store manager summary
      parameters:
        - in: query
          name: storeId
          required: true
          schema: { type: string, format: uuid }
        - in: query
          name: from
          schema: { type: string, format: date-time }
        - in: query
          name: to
          schema: { type: string, format: date-time }
      responses:
        '200':
          description: KPIs and charts data
          content:
            application/json:
              examples:
```

```
                ok:
                  value:
                    kpis:
                      totalFeedback: 245
                      avgRating: 4.2
                      positiveCount: 180
                      negativeCount: 35
                    trends:
                      - day: "2025-12-01"
                        count: 8
                        avgRating: 4.1
                      - day: "2025-12-02"
                        count: 12
                        avgRating: 4.3
  /api/reporting/distribution:
    get:
      summary: Ratings histogram
      parameters:
        - in: query
          name: storeId
          required: true
          schema: { type: string, format: uuid }
      responses:
        '200':
          description: Histogram bins
          content:
            application/json:
              examples:
                ok:
                  value:
                    bins:
                      - rating: 1
                        count: 5
                      - rating: 2
                        count: 12
                      - rating: 3
                        count: 38
                      - rating: 4
                        count: 102
                      - rating: 5
                        count: 88
```

Catalog Service

```
openapi: 3.0.3
info: { title: Catalog Service, version: 0.2.0 }
paths:
  /api/categories:
    get:
      summary: List categories
      parameters:
        - in: query
          name: storeId
          schema: { type: string, format: uuid }
      responses:
        '200':
          description: Categories
          content:
            application/json:
              examples:
                ok:
                  value:
                    items:
                      - id: "a1b2c3d4-1111-2222-3333-444455556666"
                        name: "Checkout"
                      - id: "b2c3d4e5-7777-8888-9999-0000aaaa1111"
```

```
                      name: "Product Quality"
    post:
      summary: Create category
      requestBody:
        required: true
        content:
          application/json:
            examples:
              sample:
                value:
                  name: "Customer Service"
                  description: "Interactions with staff"
      responses:
        '201':
          description: Created
          content:
            application/json:
              examples:
                created:
                  value:
                    id: "c3d4e5f6-1234-5678-90ab-cdef12345678"
                    name: "Customer Service"
```

Analytics Service

```
openapi: 3.0.3
info: { title: Analytics Service, version: 0.2.0 }
paths:
  /api/analytics/feedback/{id}:
    get:
      summary: Get analytics for feedback
      parameters:
        - in: path
          name: id
          required: true
          schema: { type: string, format: uuid }
      responses:
        '200':
          description: Sentiment and category
          content:
            application/json:
              examples:
                ok:
                  value:
                    feedbackId: "9f4a1d3c-7b8e-4f0e-9a3c-1d3c7b8e4f0e"
                    sentimentScore: 0.73
                    sentimentLabel: "Positive"
                    categorySuggestion:
                      id: "0d9f1c0a-9b6e-4b3e-8b3c-7f1c0a9b6e4b"
                      name: "Checkout"
        '404': { description: Not found }
  /api/analytics/reprocess/{id}:
    post:
      summary: Reprocess a single feedback item
      parameters:
        - in: path
          name: id
          required: true
          schema: { type: string, format: uuid }
      responses:
        '202':
          description: Accepted
          content:
            application/json:
              examples:
                accepted:
```

```
value:
  jobId: "job-20251208-000123"
  status: "Queued"
```

Deployment Pipeline Diagram

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